Moeyersons (1988,1989) studied creep movements within a 3 m thick mobile regolith mantle on a hillslope located 2 km west of the Rwabayanga sub-catchment, at a place called Rwaza hill. He showed that fissure/sliding plane configurations, typical for landslides, could develop in the silty-clayey mantle in the region. In the Rwabayanga sub-catchment we have delineated several areas where we believe that soil creep is taking place. This inventory of the hillslopes affected by creep is certainly not comprehensive and care must be taken with our interpretation. However, we believe that it is important to stress that issue as the effect of such a process could be pervasive and in the long run cause damage to infrastructures.

Creep sediment accumulation and associated landslides: Through the volume of weathered material it displaces, soil creep can contribute significantly to the formation of colluvium deposits (Thomas, 1994 – and Text Box 1). When colluvium accumulates on slopes, it can become the preferred site for landslides, as observed in neighbour regions is Uganda (Jacobs et al., 2017; Nseka et al., 2019). In the example highlighted in Figure 2 e, we identify a landslide that could be associated with this creep-related process.

Gully erosion: The conditions of mobile regolith and colluvium accumulation are also favorable to the formation of gullies (Mackey and Roering, 2011; Migoń, 2013). In some cases, also occurring as a consequence of landsliding, the interactions between landslides and gullies are typically complex and sometimes involve self-reinforcing feedbacks (Mackey and Roering, 2011; Migoń, 2013). Large gully erosion systems in which landslides are present are identified, for example, in Bujumbura in the neighbour Burundi (Dewitte et al., 2021). The origin of these large gullies is assumed to be partly associated with urban infrastructures, especially with the road network (as observed in other urban environments such as in Kinshasa, Makanzu Imwangana et al., 2014). Here we have mapped a few gullies that present such anthropogenic characteristics (Figure 2 f). Here also we must stress that the gully inventory is certainly not comprehensive. Similarly to the detection of shallow landslides, gullies can be unnoticed at the resolution of the satellite images.

Roads and trail networks: Similarly to what is observed in other mountainous regions, the frequency of landslide occurrence in Rwanda is highly increased along the roads where inadequate drainage systems, hillslope undercutting, overloading and landfills are common (Dewitte et al., 2021). These human-induced changes alter environment where landslides occur. Usually rather small landslides, often rockfall, are observed; especially during the wet seasons. Most of these landslides along the roads are usually too small to be identified in Google Earth. In this study, we were able to delineate only a few road fillings. Despite that, we can easily assume that road and trails play a role in the occurrence of landsliding.